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Tackling Global Issues: Change

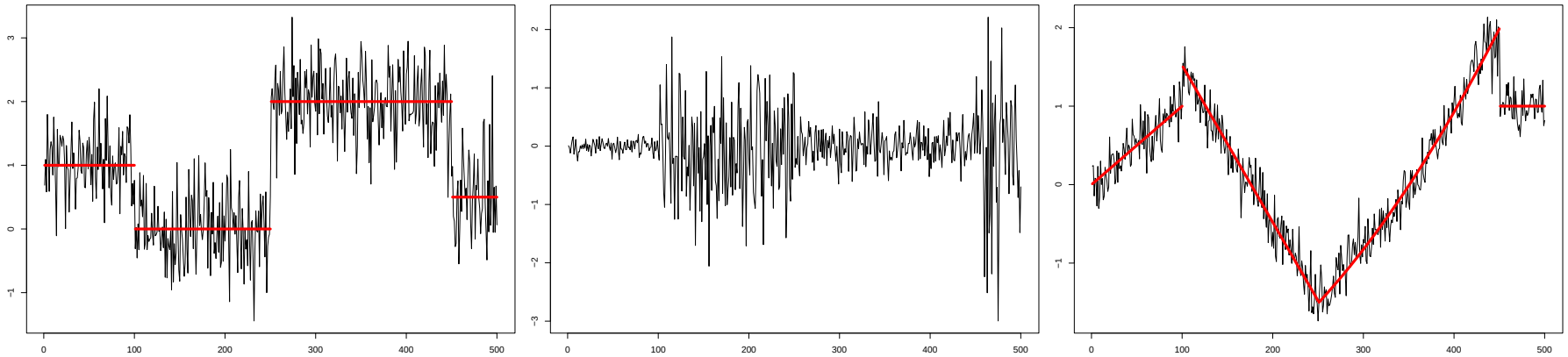
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What are changepoints?

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For data $y_1, \ldots, y_n$, if a changepoint exists at τ, then $y_1, \ldots, y_\tau$ differ from $y_{\tau+1}, \ldots, y_n$ in some way.

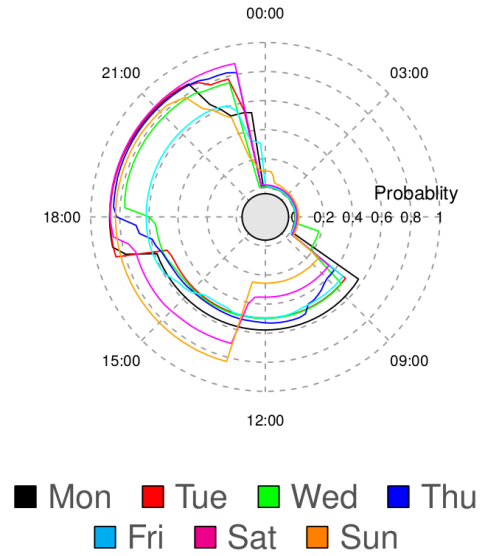
There are many different types of change.

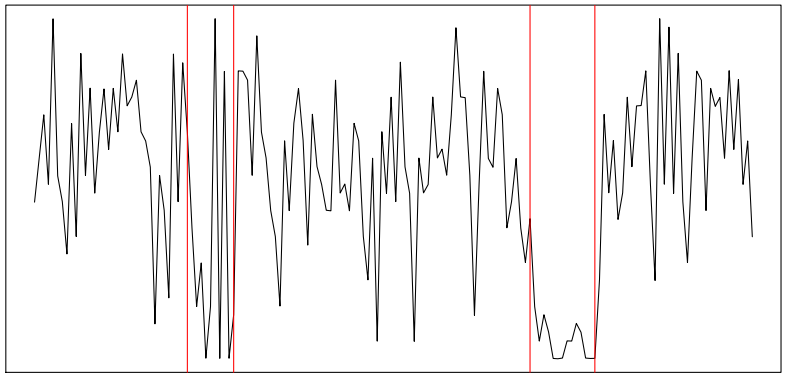


- I create new methodology and computational approaches to identify changes in complex data structures motivated by real world problems.

Aging

- Passively monitor patients at home
- Desire to have early intervention before crisis point is reached

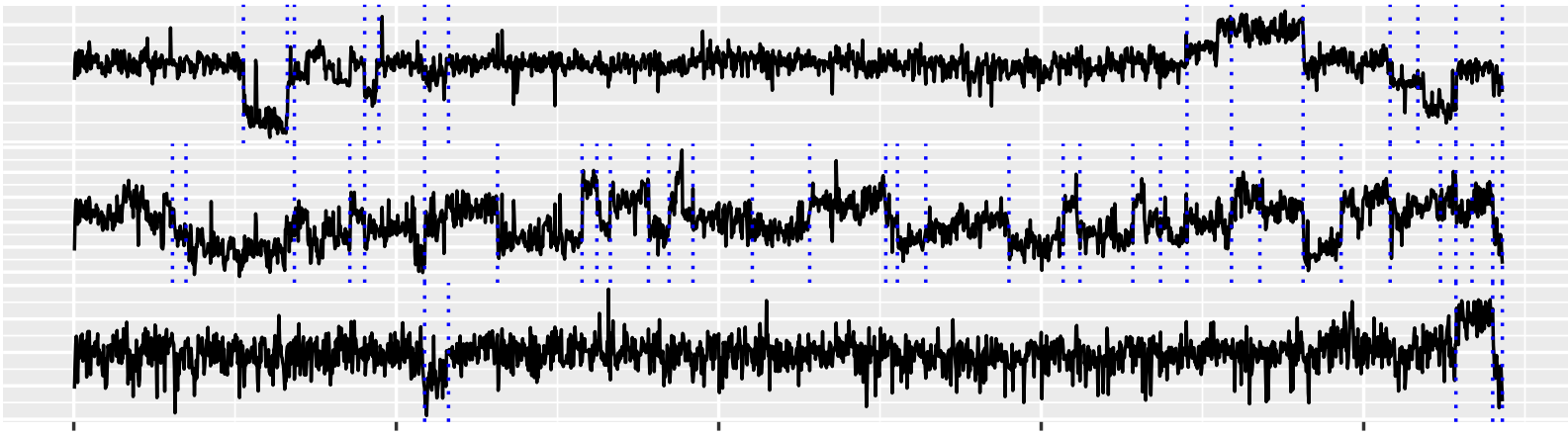




- View data as circular, e.g., each day.
- Developed circular changepoint detection to detect changes in activity levels within a day
- Changes in the detected routine signal early intervention

Health

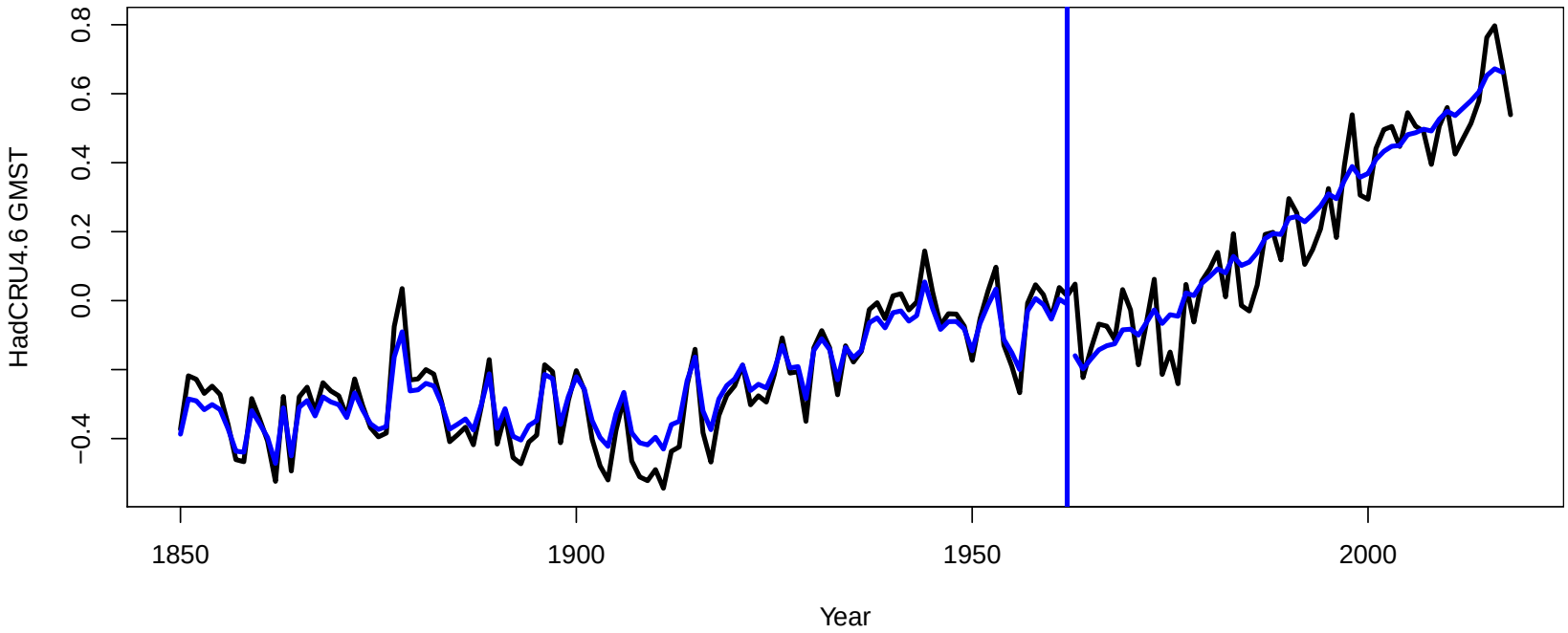
- Which genomic markers could be indications for disease?
- Only markers present in lots of patients are worth investigating.



- Developed computationally efficient subset multivariate changepoint methodology.
- Only 10 changes are present in more than 60% of the patients.
- Over 50 detected by competitors.

Climate Change

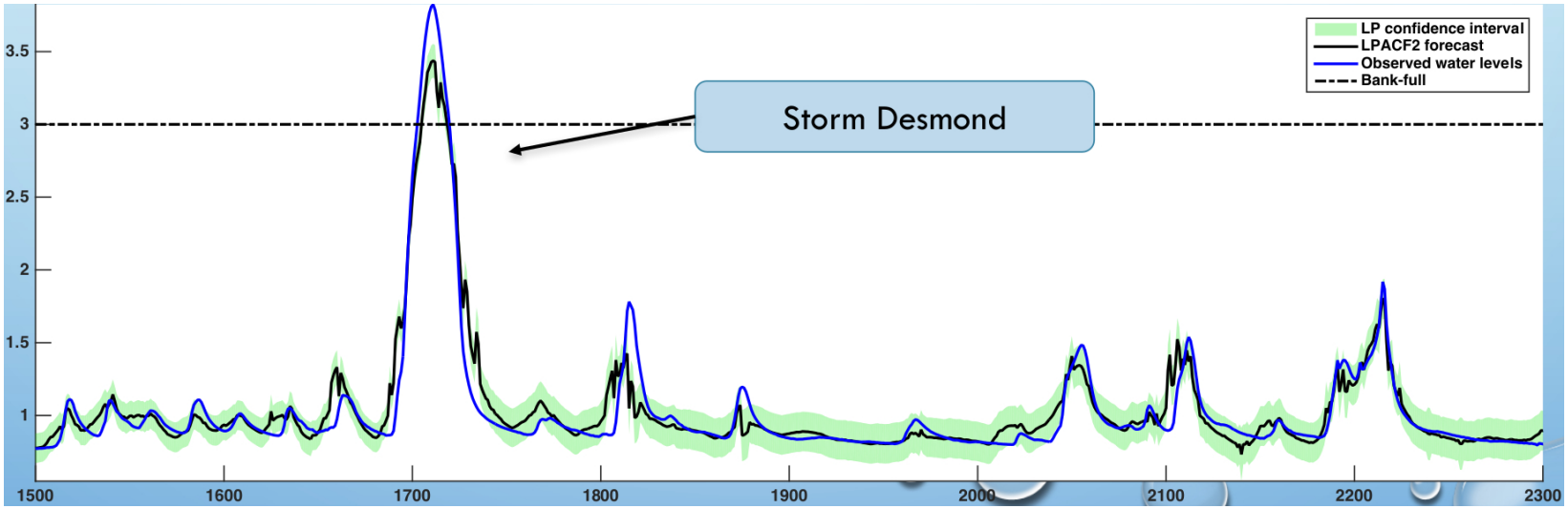
- Has there been a recent hiatus in global warming?
- Analysed several GMST time series estimates.



- Developed automatic model selection including trend, autocorrelation and changepoint detection
- This indicates no hiatus in all series.

Water

- Physical models still fail to predict the timing and extent of flooding.
- Ideally want to explain as much as we can by physical models and use statistical models to “mop up” the rest.



- Coupling the two gives better estimates and more realistic confidence levels.

Summary

- Changepoints occur in most real life data
- Detecting and documenting them improves analyses
- Multivariate, online, big data isn't a problem
- I enjoy working with messy real life data . . .
- . . . as often it sparks my next research challenge.